

Ahmed Ikram | AI Engineering

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Final-year BSc Computer Science (Data Science & AI) student at the University of Dundee (expected First-Class, 2027). Builds production grade LLM systems with the evaluation harnesses that prove they work: **RAGAS**-scored retrieval, **QLoRA** execution-based fine-tuning, statistically-gated promotion, and vLLM. An **MCP-served RAG agent** roughly doubles retrieval recall over deterministic tool routing (0.708 vs 0.350); a QLoRA champion with 7.0x the base model on real SWE-bench tests. Seeking an **AI Engineering** role or Applied ML / MLOps position.

EDUCATION

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- On track for a **First-Class Honours** degree, Graduation: **June 2027**

SKILLS

LLMs & Generative AI: PyTorch · Hugging Face Transformers · QLoRA/LoRA/PEFT · TRL · Unsloth · vLLM · Prompt & Context Engineering

AI Systems & Agents: LangGraph · LangChain · Model Context Protocol (MCP) · AWS Bedrock · ChromaDB · Cross-Encoder Reranking · Multimodal (Vision-Language) Extraction · LLM Guardrails

Classical & Applied ML: XGBoost · Isolation Forest · scikit-learn · Optuna · LSTM Forecasting

Evaluation & Observability: RAGAS · Execution-Based & LLM-as-Judge Evaluation · LangSmith · Langfuse · Evidently AI

MLOps & Serving: MLflow · Weights & Biases · Airflow · FastAPI · Modal · AWS SageMaker

Languages: Python · SQL · TypeScript · JavaScript · Bash · Rust / PyO3

Data & Backend: PostgreSQL · Redis · Kafka · Pandas · NumPy · Pydantic

Cloud & Infrastructure: AWS · Google Cloud · Terraform · Docker · GitHub Actions · Linux

PROJECTS

SWE-Qwen Automated LLM Fine-Tuning, Evaluation & Serving Platform

Python · PyTorch · QLoRA · Unsloth · Hugging Face · Modal · vLLM · Weights & Biases · Langfuse · Terraform · Google Cloud

- Model-agnostic LLMops platform spanning schema-validated data preparation, QLoRA fine-tuning, execution-based evaluation and statistically-gated champion/challenger promotion, serving the winner as a config-driven OpenAI compatible API - versioned GCS/W&B artifacts shared across every stage.
- Six-stage data pipeline turned 20,477 raw SWE-bench issues into a leak-free 17,456-example corpus across 46 repos; evaluation materialises each repo in its official Docker image and runs its real pass/fail test suite for ground-truth, execution-based scoring.
- Scored 3 QLoRA Qwen3-14B variants (4-bit NF4, Unsloth + FlashAttention on Modal A100); champion reaches 17.20% fail-to-pass - 7.0× base (95% CI 11.1-25.8%, McNemar $p < 1e-6$) - and 90.10% unrelated pass-pass rate (+61.6 pt vs base); Promotion needs a 4-condition statistical gate so a challenger can never self-certify a win.
- Served on Modal scale-to-zero with per-request LoRA swapping via vLLM+AWQ (\$0 idle); a 20-instance F2P smoke gate (SMOKE_TOLERANCE=0.05) blocks regressing PRs. GCP via Terraform + Workload Identity Federation; 1,456 tests.

ATM Log Aggregation, Anomaly Detection & Diagnostics Platform

LangGraph · MCP · ChromaDB · Ollama · FastAPI · XGBoost · MLflow · Kafka · Redis · Docker · Terraform · AWS

- ML and Backend lead on a 7-person Agile team for NCR Atleos: Independently built the ML anomaly detection engine, agentic hybrid RAG assistant, AWS infrastructure, and 1,438-test suite as a post-deadline extension on the team-delivered foundation.
- Hybrid RAG as a LangGraph agent over a 12-tool MCP server, semantic chunking + local 'nomic-embed-text' embeddings, cross-encoder reranking, grounding-gated re-retrieval, then reflexion / self-consistency / citation checks fused with Platt-calibrated confidence (ECE gate 0.10).
- RAGAS evaluation across 3 retrieval strategies (50-query golden set): adaptive routing with re-retrieval **nearly doubles** context recall over deterministic selection (0.708 vs 0.350) at the best context precision (0.874); a CI gate blocks any merge regressing a RAGAS metric by >0.05 .

- 3-layer anomaly detection (XGBoost + Isolation Forest ensemble, Z-score & 7-heuristics engine) runs every 30s on a live Kafka stream, reaching 0.70 F1 at 97.3% precision via a 200-point Youden sweep, versioned in MLflow with champion aliases and cross-checked against a live SageMaker endpoint.
- Terraform-provisioned AWS (ECS Fargate, RDS, S3, SageMaker, Secrets Manager) via OIDC-authenticated CI/CD, validated by 1,438 tests.

StockLens FinTech Spending & Investment App

LangGraph · AWS · PyTorch · Optuna · Rust / PyO3 · Airflow · MLflow · Evidently AI · Tesseract · PostgreSQL · Redis · Terraform

- 2-node LangGraph ReAct agent (explicit StateGraph, not the `create\_react\_agent` wrapper) exposes 16 tools to AWS Bedrock with SSE-streamed responses and dual Redis/PostgreSQL persistence surviving restarts; correctness judged in CI by an LLM-as-judge with 10%-sampled LangSmith tracing.
- Confidence-gated 4-stage receipt-extraction cascade (Tesseract OCR → Bedrock Vision-LLM → Text fallback → Degraded Fallback) with rapidfuzz merchant matching, escalating only when confidence drops below thresholds.
- Global LSTM (entity embeddings, Optuna 50-trial HPO, focal loss) reaches 51.6% 5-day directional accuracy vs a 33% 3-class baseline (0.97 simulated Sharpe). 17 features computed in a Rust/PyO3 engine; retrained weekly via Airflow with Evidently drift monitoring (PSI/KS/JSD) and MLflow champion/challenger promotion at >2 pp.
- 90% test coverage gate on Terraform-provisioned AWS (ECS Fargate, EFS champion serving); 9-parallel CI jobs and a 7-stage OIDC-authenticated CD.

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	Feb 2025
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	Oct 2024
<u>AWS Academy Graduate - Cloud Foundations</u>	Mar 2024

EXPERIENCE

Beaully Mini Market , Dundee, Scotland	
<i>Retail Assistant</i>	<i>September 2023 - Present</i>
Zulus Peri Peri Grill , Dundee, Scotland	
<i>Front of House Worker</i>	<i>June 2021 - September 2023</i>